Data-Driven Customer Segmentation for Targeted Marketing Strategy

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DECLARATION OF AI USAGE

Artificial Intelligence Declaration

I confirm that this report has been prepared in accordance with the University's Academic Integrity Policy regarding the use of Generative Artificial Intelligence tools.

AI Tools Used: Perplexity, Notebook lm and ChatGPT were used for code debugging, structuring report sections, and proofreading.

Extent of AI Usage:
Generative AI was used to support data visualization, code debugging, structuring report sections, and proofreading.

Report Structure

1.Executive Summary

This analysis segments the retailer's loyalty customer base (n=3,000) using six months of transactional basket behaviour and category-level spend to support targeted marketing and retention decisions.

A key data-quality issue was addressed before modelling, bakery spend was recorded as zero in the category spend table. So, Bakery spend was recalculated from line-item transactions and overwritten, affecting **98.8% of customers (2963/3000)** and materially improving feature validity.

After cleaning and transforming the data into a customer-level feature set (frequency, spend intensity, basket composition, recency and category preferences), then applied z-score scaling and performed PCA to reduce noise and multicollinearity. KMeans clustering was then evaluated across k=4–10 using elbow and silhouette diagnostics, with **k=5** selected because it produced the strongest silhouette score within that range (**0.171**) alongside other usable segment sizes.

Key differences across segments include **visit frequency**, **basket spend/quantity**, **recency**, and **concentration of spend in high-signal categories** (e.g., tobacco/lottery/cashpoint vs. broader grocery mix).

The resulting five segments exhibit distinct shopping patterns and value profiles:

1.A large low–mid value group with lower engagement.

2.A small high value "premium heavy shopper" group with large baskets and broad category breadth.

3.Hyper-frequent top-up shoppers with elevated tobacco/ancillary behaviour.

4. Frequent essentials shoppers suitable for basket-building.

5. "Stock-up" shoppers with higher basket value but less recent purchasing

Recommend prioritising two target segments:

1.Customer clutter 2: Should be protected and grown through retention-oriented tactics (VIP perks, personalised bundles, service/availability focus).

2.Customer clutter 5: It offers strong upside through reactivation and basket-building (timed triggers based on recency, mission-based multi-buy offers across fresh/bakery/essentials).

Together these actions balance near-term revenue impact with scalable targeting. Overall, the segmentation provides an actionable view of the customer base and a foundation for targeted promotions, retention interventions, and future uplift modelling.

2.Feature Description & Data Preparation

2.1 Data sources and customer-level build

The dataset included four tables describing the same 3,000 loyalty customers over a six-month period: a customer summary (customers_sample), visit-level baskets (baskets_sample), category spends (category_spends_sample), and line-items (lineitems_sample).
To make the data usable for clustering, single modelling table was created with **one row per customer** by aggregating the basket table to customer level (frequency, spend intensity, basket size, and temporal engagement) and then joining category-level spends on the same customer identifier. These features combine define overall behaviour of customers.

2.2 Data preparation and cleaning decisions

To cluster properly data needs to be free of inconsistent scales or recording mistakes, as it is distance based and can overwhelm true behavioural differences, we want model to learn. Though dataset was **not containing any null values,** there were inconsistencies and negative values in datasets.

Currency formats and data types → converted to numeric floats:

Several spend fields were stored as currency strings (e.g., "£0.00"). These were converted to numeric floats so they can be meaningfully compared and scaled. If values remain as strings, they either fail to process or get implicitly mishandled. Converting to floats preserves the true monetary magnitude and allows consistent comparison across customers.

Negative spends → treated as data issues and were replace with 0:

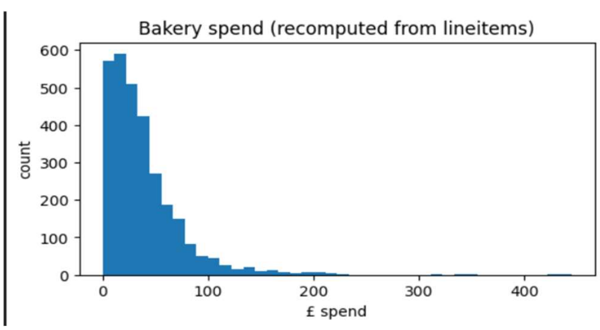
Negative numbers typically reflect **returns, corrections, or recording adjustments**, and if left unchanged they can distort distances, pull centroids, and create clusters that reflect accounting artefacts rather than behaviour. So, negative values were replaced to 0 rather than deleting those rows because there would have been loss of customer data. Dropping those customers would bias the sample toward frequent shoppers.Cleaning them keeps the features aligned to the business meaning and makes segment profiles interpretable for marketing.

2.3 The Bakery correction

A known issue in the dataset is that bakery spend was recorded as zero for all 3,000 customers in category_spends column, even though bakery purchases exist in lineitems_sample.

To ensure the segments reflected real diversity in purchasing behaviour, lineitems_sample was filtered to category == 'BAKERY', summing spend per customer, and overwriting the bakery column in the category spend table. This changed bakery values for **2,963 out of 3,000** customers **(98.8%) and** shifted the bakery mean from **£0.00 to £38.2.** So, it's not just cleanup, it meaningfully impacts the segmentation input data.

Index	customer_number	bakery_original	bakery (Corrected)
1573	11401	0.0	444.37
1538	13793	0.0	430.50
963	2067	0.0	353.05
1166	12221	0.0	335.01
1765	11511	0.0	314.28



2.4 Feature engineering

To keep segmentation grounded in Spend, Frequency, and shopping behaviour, features were organised into four complementary feature blocks to capture value, engagement, visit structure, and product preference.

A) Frequency & engagement (how often and how recently they shop)

- **n_baskets**: number of store visits during the period — it shows customer frequency and loyalty.
- **recency_days**: days since the last recorded visit — shows customers current engagement and separating active shoppers from those beginning to drop off.

B) Basket value / spend intensity (how much they spend per visit)

- **total_basket_spend**: overall customer value across the period.
- **avg_basket_spend**: average spend per visit.
- **max_basket_spend**: highlights customers who sometimes make large payments.
- **std_basket_spend**: distinguishes consistent spending patterns from irregular spikes and dips.

C) Basket size & mission breadth (what a visit looks like)

- **total_basket_qty, avg_basket_qty, std_basket_qty**: Capture purchase volume patterns, separating "small, frequent top-ups" from "larger, less frequent restocks," and whether that volume is consistent or fluctuating.
- **avg_basket_categories, std_basket_categories**: These measures tell you if a customer usually buys from one category or many categories, and whether their visits stay similar each time or change a lot.

D) Category preference (what they tend to buy)

- Category-level spend across approx.20 categories (e.g., **fruit & veg, dairy, grocery food, tobacco, drinks, lottery, cashpoint**, etc.), including the **corrected bakery spend** derived from line items. This block is the most directly actionable for marketing because it links each segment to clear product interests and promotion triggers.

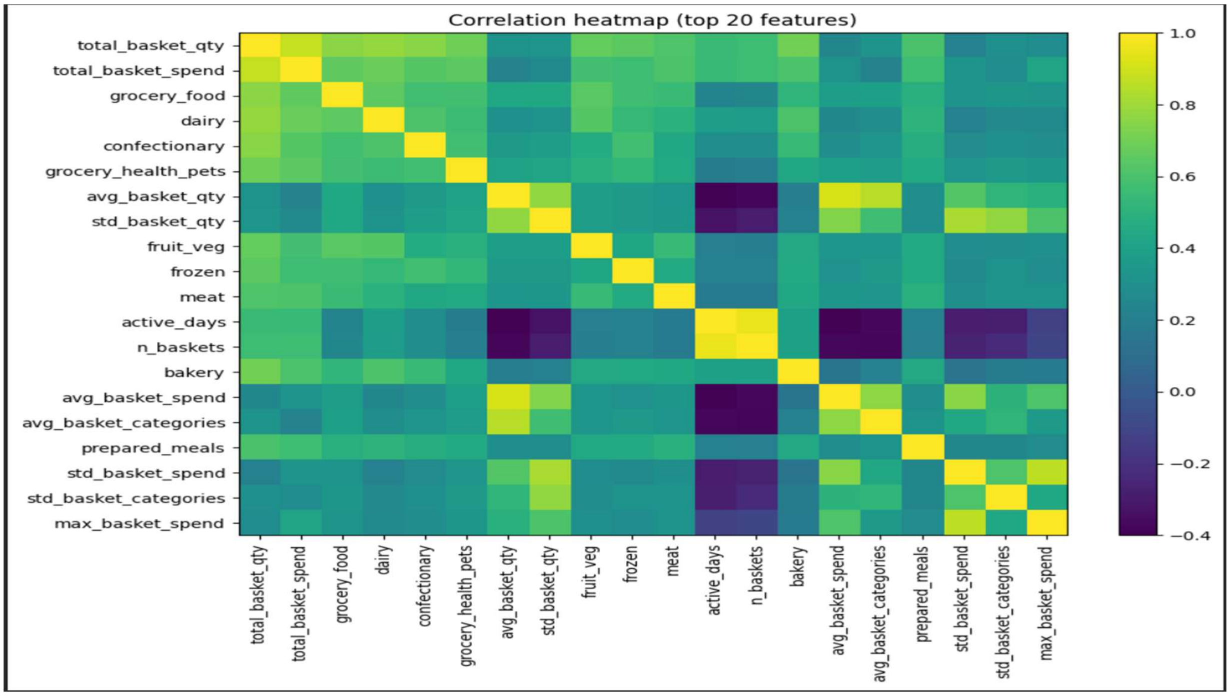
Together, these blocks provide a complete picture of customer behavioural profile. They measure not only **"how valuable"** a customer is, but also **"how they shop"** (often small trips vs occasional large shops, steady vs irregular patterns) and **"what they buy"** (category mix). That combination is what makes segments interpretable and usable to target with marketing actions.

2.5 Avoiding "double counting" before PCA

Clustering can be skewed when multiple variables contain same information which are highly correlated inputs. Correlation is problematic because it effectively double counts the same behavioural signal and can dominate distance calculations.

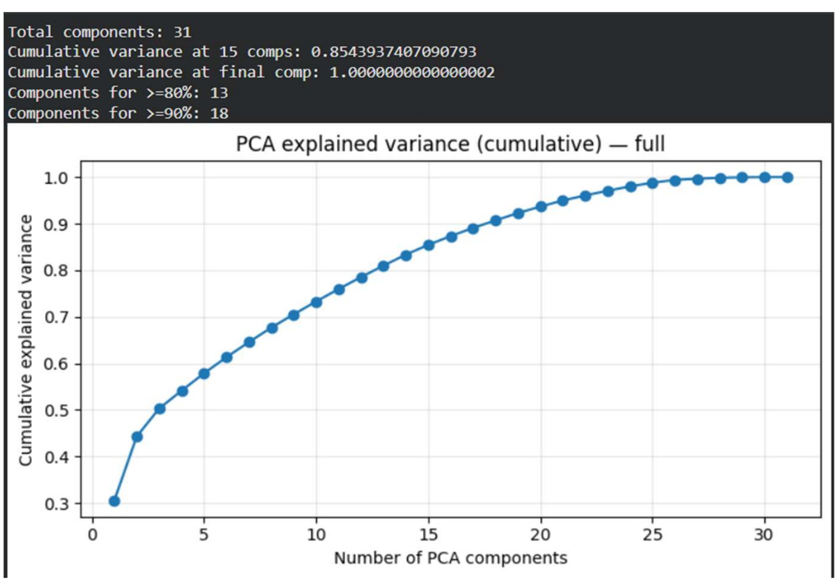
To avoid this, strongly correlated features were removed before running PCA, rather than relying on PCA to resolve all redundancy on its own.

To that we first dropped obvious duplicate summary fields (multiple versions of basket aggregates capturing the same behaviour). We then ran a correlation screen using a strict threshold **($|r| > 0.95$)** to identify near-identical features. This process flagged variables such as **tenure_days** and **active_days**, which overlap strongly with other engagement measures. These are removed while retaining more directly interpretable behavioural indicators **recency_days** (how recently the customer visited) and **n_baskets** (how often they visited) because they translate cleanly into marketing actions and segment narratives.



2.6 Scaling and PCA (dimensionality reduction)

All features were **standardised (z-scores)** before PCA and k-means so that large spend variables do not dominate smaller behavioural measures. **PCA** was then used to reduce dimensionality while retaining most behavioural signal. The cumulative explained-variance plot shows **31 components** with variance accumulating gradually rather than concentrating in a few factors. For clustering, we used **13 components (~80% variance)** to keep the model compact without over-compressing the data; **18 components** would be required to reach **~90%**.



3. Customer Base Summary (EDA)

An overall view of the engineered behavioural features (frequency, spend, basket size, basket breadth, and recency) suggests the customer base is a blend of many shoppers appear to use the store for frequent, smaller "top-up" visits, while a meaningful share make less frequent but larger "mission" baskets.

3.1 What the average customer looks like

Looking across the engineered behavioural features (frequency, spend, basket size, basket breadth, and recency), the customer base appears to combine two common convenience-retail patterns: frequent low-value **top-up trips** and less frequent, higher-value **mission shops.**

Using the customer-level basket features:

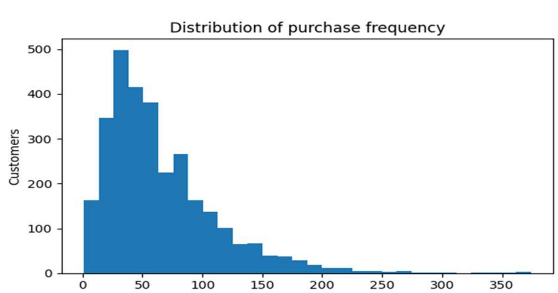
- Customers make **around 65 visits (baskets)** on average.
- The mean **recency is ~8 days** since the last recorded shop, indicating generally active engagement.
- Average basket spend is **~£14.8**, while average total spend across the period is **~£770 per customer**.
- A typical basket contains **~11 items** and spans **~4–5 categories**, suggesting moderately broad baskets rather than single-item purchases.

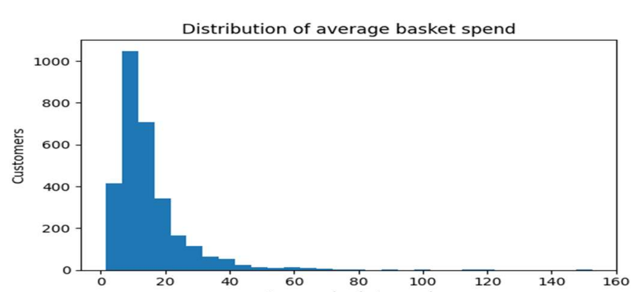
3.2 Distribution patterns

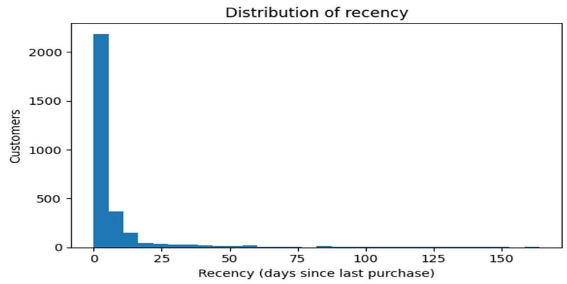
- **Frequency and spend are unevenly distributed.** A substantial portion of customers shop only occasionally, while a smaller subset visits very frequently. This creates naturally different engagement profiles that are difficult to address with one generic marketing approach.
- **Spend measures strongly show a right skewed.** As is common in retail, a relatively small group of customers contributes a disproportionately large share of revenue. This is exactly where segmentation becomes valuable, it helps marketing identify which groups to prioritise and how to tailor offers rather than spreading budget evenly.
- **Recency separates "active" from "at risk."** Customers who have shopped recently are likely still engaged, whereas those with longer gaps may require reactivation messaging. Including recency ensures segments capture current engagement, not just historical spend.

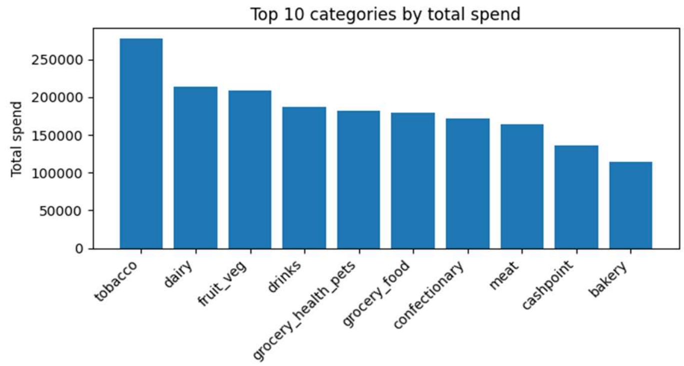
3.3 Category spend overview (what people buy)

Category totals show which departments account for the largest share of revenue overall**,** while the per-customer distributions highlight which categories are "broadly common" vs. those that differentiate niche behaviours (e.g., tobacco/lottery/cashpoint/seasonal gifting). This overview sets up pen portraits, where the segments are described not just by value, but by shopping mission and product interest.









4. Segmentation Methodology

K-Means was used because it is a widely adopted, computationally efficient method for partitioning customers into similarity-based groups. It iterates between **assigning each customer to the nearest cluster centre (centroid)** and **updating centroids to the mean position of their assigned customers**. Because it is distance-based, results depend on **proper scaling and well-designed behavioural features**, otherwise, high-magnitude variables (e.g., total spend) can dominate distances and distort the clustering.

4.2 Pre-processing pipeline before clustering

Before applying k-means, a structured pre-processing pipeline was used to ensure that distances between customers reflected behaviour rather than artefacts of scale or redundancy.

Standardisation (z-scores): All features were standardised so spend and count-based measures (e.g., baskets, quantities) contribute comparably to distances, preventing large spend variables from dominating and meeting the specification's expectation to cleanse/normalise data before modelling.

Correlation control: Clustering can be distorted if several inputs measure the same underlying behaviour. To avoid this "double counting" effect, highly correlated features were screened and redundant variables removed **before** dimensionality reduction, rather than relying on PCA to resolve all redundancy automatically.

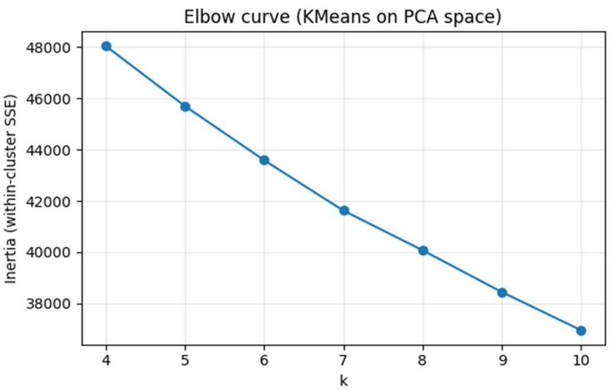
PCA for dimensionality reduction: PCA was applied to compress the feature space, reduce residual collinearity and noise, while retaining most behavioural signal. The explained-variance curve shows **13 components capture ~80%** (and **18 ~90%**) of variance. Clustering was run in PCA space to create a cleaner distance structure and a more stable segmentation.

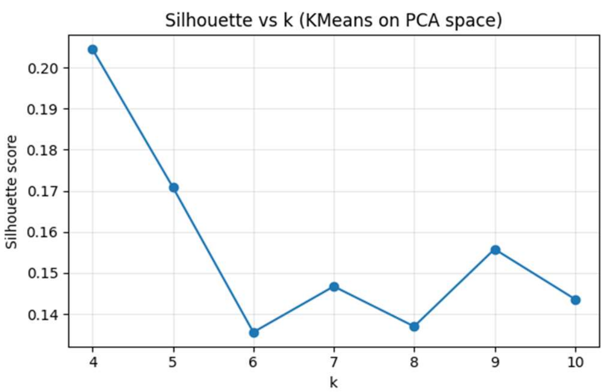
4.3 Choosing the number of segments (k)

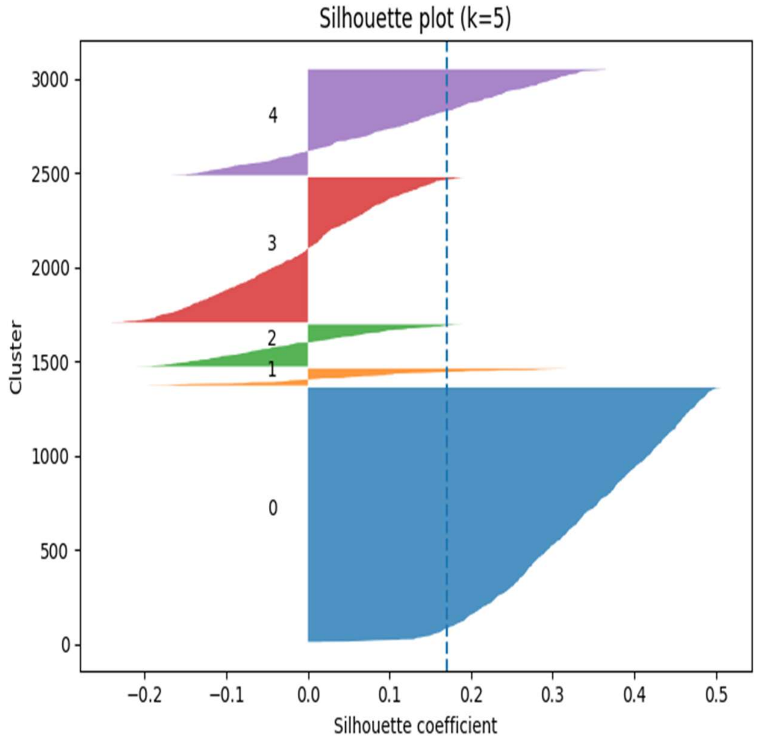
To choose k, solutions from k = 4 to k = 10 were compared using two standard diagnostics:

- **Elbow method (inertia / within-cluster SSE):** It tracks how fast within-cluster variation falls as k increases; a clear "bend" signals diminishing returns.
- **Silhouette score and silhouette plot:** Measures how well each customer fits their cluster versus the nearest alternative; higher is better separation, but interpret the score alongside the silhouette plot (not as a single decisive number).

Within the required **5–7** range, **k = 5** gave the best balance: it had the highest silhouette score (**0.171**) and produced segment sizes suitable for marketing. Although the silhouette is not "high" (retail behaviours overlap rather than separate cleanly), quality was assessed using the silhouette diagnostics together with the clarity and distinctiveness of the segment profiles, not the score alone.







5. Analysis & Description of Results

5.1 Segment sizes and overall shape

Segment sizes are uneven because many customers behave broadly "average," while smaller groups show sharper patterns (very high value, very high frequency, or distinctive category missions). **Customer clutter 1** is the mainstream base (**45.0%, n=1,351**) and matters mainly due to scale, while **Customer clutter 2** is small (**3.0%, n=91**) but clearly high value and therefore disproportionately important. The remaining segments support targeting: **Customer clutter 4** is large and highly active (**25.7%, n=771**), **Customer clutter 5** reflects a stock-up pattern (**18.8%, n=563**), and **Customer clutter 3** is smaller but distinctive (**7.5%, n=224**) with very high visit frequency and strong category signals.

Index	Segment Name	n_customers	pct_customers (%)
0	Customer clutter 1	1,351	45.0%
1	Customer clutter 4	771	25.7%
2	Customer clutter 5	563	18.8%
3	Customer clutter 3	224	7.5%
4	Customer clutter 2	91	3.0%

5.2 Statistical summaries

Cluster differences were summarised using groupby() means and medians in the **original feature space**, enabling direct comparisons in business terms (visits, recency, basket spend/size, category spend) without interpreting PCA components. This keeps profiles clear and actionable for decision-makers. For readability, the main report includes one compact KPI table (Table 5.2), while full mean/median profiles for all engineered features are provided in the appendix and referenced for completeness.

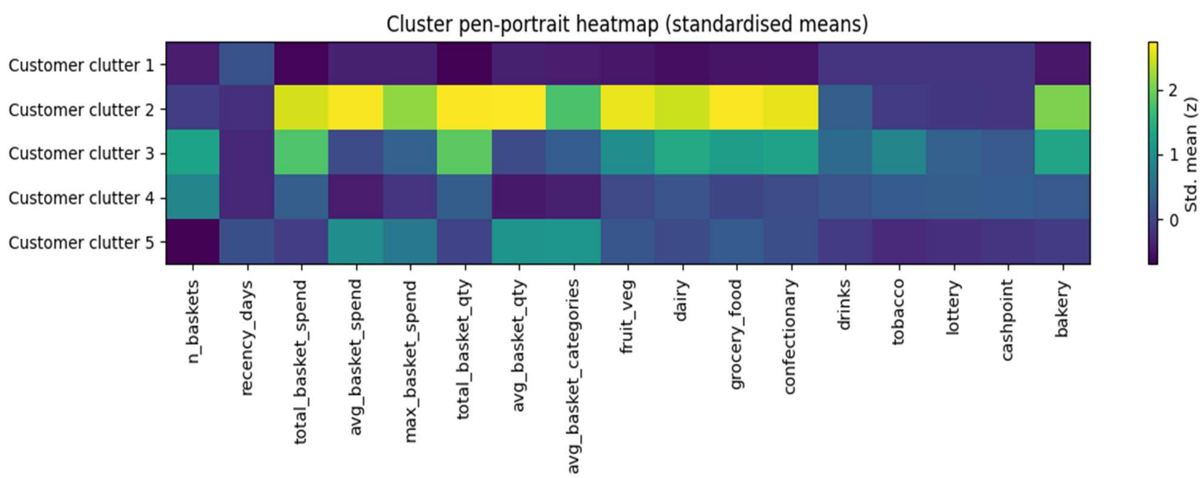
Table 5.2 – Pen portrait KPIs (mean values by segment) highlights four clear patterns:

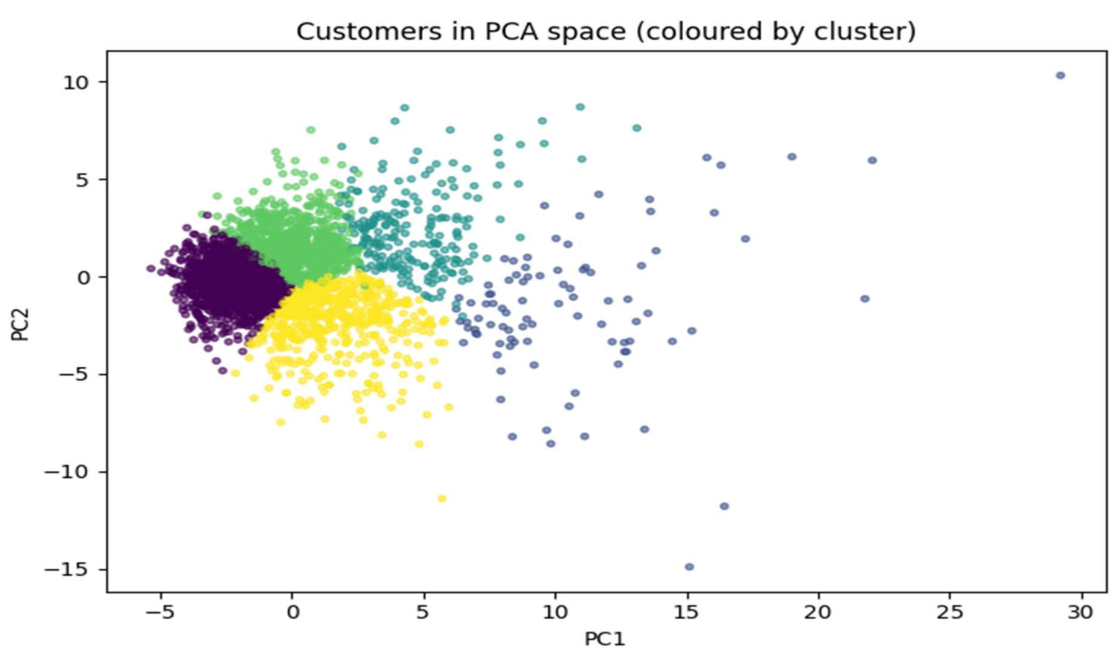
Value gradient: mean total spend rises steadily from Customer clutter 1 (£415) → Customer clutter 5 (£741) → Customer clutter 4 (£963) → Customer clutter 3 (£1,756) → Customer clutter 2 (£2,158).

Engagement split: Customer clutter 3 and 4 are the most active and most recent shoppers (high basket counts and ~2-day recency), whereas Customer clutter 1 and 5 are less recent on average (~11 days), indicating potential for reactivation.

Shop size and breadth: Customer clutter 2 and 5 have larger shopping, reflected in higher average basket spend and quantity, alongside broader category coverage per trip.

Distinctive category signals: Tobacco spend is particularly higher in Customer clutter 3 and 4, while Customer clutter 2 shows the strongest bakery spend, reinforcing that segments differ not only in value but also in purchasing mix.





5.3 Pen profiles

Customer clutter 1 — "The quiet majority"

This is the largest segment (**45%**) and represents broadly "average" shoppers who do not stand out on value or engagement. They have the lowest **average basket spend (£10.63)** and **total spend (~£415)**, and while not inactive, they are less fresh than the frequent segments (**recency ~11.8 days**). Overall, they resemble a background population: many small-to-medium trips with no strong signature category pattern. Any targeting should be **low-cost and scalable** (mass offers, app pushes, simple basket builders) because the upside per customer is limited.

Customer clutter 2 — "Premium mission shoppers"

This is a small segment (3%), but they're the clear outliers on value. They have the highest total spend (**~£2,158**) and the highest average spend per basket (**£44.79**), with very large baskets (avg quantity **34.6**) and wide category breadth (avg **7.63** categories per trip). Their recency is also relatively strong (**~3.5 days**), which suggests they are still actively engaged. The "personality" here is the planned, high-ticket shopper: fewer wasted trips, bigger missions, and a lot of money left on the table if they churn.

Customer clutter 3 — "Everyday top-up regulars with strong tobacco/ancillary behaviour"

This group shops constantly (mean **127 baskets**) and is very recent (**~1.9 days**), but their baskets aren't especially expensive (avg spend **£15.78**). What makes them distinctive is the heavy tobacco signal (**~£268** mean), plus higher lottery and cashpoint compared with most other segments. They read like convenience-driven shoppers: lots of quick trips, habitual behaviour, and a high reliance on a few categories. If the store annoys them (out-of-stock, queues), they're the kind who switch habits fast.

Customer clutter 4 — "Habitual micro-trip shoppers"

Customer clutter 4 is large (25.7%) and extremely active (recency **~1.9 days**, **106 baskets**), but their average basket spend is the lowest of all segments (**£10.04**). They look like people who pop in frequently for small baskets — not big weekly missions. They also show elevated tobacco/lottery/cashpoint signals (though not as extreme as Customer clutter 3). This is a classic "high footfall, low basket" segment: the win isn't getting them to visit more, it's getting them to add one or two extra items per trip.

Customer clutter 5 — "Stock-up shoppers (big baskets, less recent)"

This segment is moderately sized (18.8%) and behaves very differently from the frequent groups. They have fewer visits (mean **32 baskets**) but much larger baskets (avg spend **£26.11**, avg quantity **20.5**) and broad category mix (avg **6.56** categories). Their recency is higher (**~11.0 days**), which makes them interesting: they're valuable when they show up, but there's a clear reactivation lever if you can time offers around their purchase cycle. The "personality" is the planned stock-up mission: fewer trips, heavier baskets, and broad shopping lists.

6. Summary & Strategic Recommendations

6.1 Summary of what the segmentation shows

The five segments provide a coherent view of the customer base, from a large mainstream low-spend group to smaller segments with clearer shopping missions (high-value stock-up trips, highly frequent top-ups, and a premium high-ticket segment). This matters because one marketing message will not perform equally across these behaviours; segmentation reduces wasted spend and improves conversion by matching offers to mission and value.

The strongest differentiators are: **i]value** (total and average basket spend), **ii]engagement** (frequency and recency), and **iii] mission/category signals** (e.g., tobacco/ancillary vs broader grocery missions) — the same behavioural dimensions highlighted by the client for marketing decision-making.

6.2 Business case

From a commercial perspective, the segmentation provides practical prioritisation of marketing effort. It helps identify where to focus retention for high-value customers, where to grow basket size among frequent low-value trip shoppers, and where to run reactivation activity for customers whose patterns suggest periodic stock-up cycles. This aligns the targeting stage of segmentation, assess segment attractiveness and focus resources on groups the organisation can reach and influence effectively.

6.3 Recommended target segments

1. **Customer clutter 2 (Premium heavy shoppers, 3.0%)**: This segment should be prioritised for retention and value protection. It has the highest spend intensity (mean total spend **£2157**; mean average basket spend **£44.79**) and broad shopping missions (mean **7.63** categories per basket).

   Recommended actions: VIP-style benefits, personalised bundles, and service/availability interventions aimed at preventing churn and increasing share-of-wallet.

2. **Customer clutter 5 (Stock-up shoppers, 18.8%):** This segment should be prioritised for reactivation and basket-building. This segment has high basket value (mean average basket spend **£26.11**) and high basket quantities with broad category mix, but less recent activity (mean recency **11.03 days**).

   Recommended actions: Timed triggers based on recency (e.g., "it's time to restock" offers), multi-buy offers aligned to increase shopping volume, and cross-category bundles (fresh + bakery + grocery staples) to increase trip conversion and average order value without training the segment to wait for deep discounts

7. Appendix

A1 Full Feature Means by Segment (Table 5.2)

Mean profile:

segment_name	n_baskets	recency_days	total_basket_spend	avg_basket_spend	max_basket_spend	total_basket_qty	avg_basket_qty	avg_basket_categories	fruit_veg	dairy
Customer clutter 1	45.34	11.80	414.65	10.63	32.11	310.52	7.91	4.00	36.07	38.96
Customer clutter 2	61.77	3.51	2157.55	44.79	112.04	1680.15	34.59	7.63	254.99	214.84
Customer clutter 3	126.75	1.91	1755.98	15.78	56.27	1338.00	12.08	5.25	139.32	150.71
Customer clutter 4	106.45	1.90	962.90	10.04	39.00	713.29	7.43	4.04	74.36	83.67
Customer clutter 5	32.33	11.03	740.82	26.11	66.01	584.64	20.52	6.56	85.08	77.18

A2 Full Feature Medians by Segment (Table 5.2)

Median profile:

segment_name	n_baskets	recency_days	total_basket_spend	avg_basket_spend	max_basket_spend	total_basket_qty	avg_basket_qty	avg_basket_categories	fruit_veg	dairy
Customer clutter 1	42.0	3.0	403.68	9.84	28.05	308.0	7.28	3.88	27.48	33.69
Customer clutter 2	48.0	1.0	1951.11	38.77	103.37	1563.0	29.16	7.36	228.87	206.10
Customer clutter 3	115.0	0.0	1598.58	14.52	51.36	1262.5	11.09	5.04	120.42	135.59
Customer clutter 4	96.0	0.0	886.56	9.31	31.92	693.0	7.17	3.95	65.15	77.20
Customer clutter 5	31.0	3.0	723.87	23.50	59.61	586.0	18.11	6.39	74.62	71.47

A3. Pen portrait table (Table 5.3)

segment_name	n_customers	pct_customers	recency_days	n_baskets	avg_basket_spend	total_basket_spend	avg_basket_qty	avg_basket_categories	bakery	tobacco	lottery
Customer clutter 1	1351	45.0	11.80	45.34	10.63	414.65	7.91	4.00	21.12	55.72	6.71
Customer clutter 2	91	3.0	3.51	61.77	44.79	2157.55	34.59	7.63	113.16	71.80	7.30
Customer clutter 3	224	7.5	1.91	126.75	15.78	1755.98	12.08	5.25	85.91	268.03	32.65
Customer clutter 4	771	25.7	1.90	106.45	10.04	962.90	7.43	4.04	47.69	150.49	30.83
Customer clutter 5	563	18.8	11.03	32.33	26.11	740.82	20.52	6.56	35.15	35.12	3.48